

Yuxiang Wan

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Education

University of Illinois at Urbana-Champaign, B.S. in Computer Engineering GPA: 4.00/4.00	Sep. 2023 – May 2027
Zhejiang University, B.E. in Electrical and Computer Engineering GPA: 3.955/4.0, Rank: 1/70	Sep. 2023 – May 2027

Research Experience & Interest

My prior research focused on **graph-structured methods for in-context learning**, aiming to improve the **reasoning and generalization** of large language models via structured knowledge integration. Building on this experience, I am now exploring **machine learning systems**, particularly **efficient inference** and scalable system support for modern ML workloads, including **interactive settings such as reinforcement learning and multi-agent learning**.

Research Intern, AI-Net Lab, Zhejiang University Advisors: Yifei Sun, Yang Yang Jun. 2024 – Apr. 2025

- Conducted a systematic literature review on in-context learning and graph-based representation learning, contributing to the formulation of a graph-structured example selection framework.
- Designed and implemented experiments: leveraged graph structures and **PageRank-based example selection** with multi-level encoding (node, path, subgraph), achieving **2.74% improvement** over SOTA
- Drafted and revised a research paper; conducted extensive experiments and analysis during rebuttal.

Projects

ECE 391: UNIX-like Operating System Kernel	<i>C, RISC-V Assembly</i>	<i>Fall 2025</i>
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- Built a UNIX-like OS kernel on RISC-V from scratch: implemented Sv39 virtual memory (3-level page tables, demand paging), process abstraction (fork, exec), and preemptive multitasking with round-robin scheduling
- Developed VIRTIO block device driver and KTFS filesystem with 64-block write-back cache, supporting CRUD operations, indirect/doubly-indirect block addressing, and persistent storage
- Implemented 15+ system calls (fork, exec, read, write, pipe, etc.) and a functional shell with I/O redirection and piping; wrote Unix utilities (cat, ls, wc, xargs) for command-line interaction

Selected Honors and Awards

National Scholarship , Ministry of Education of China (Top 1%)	2024, 2025
Dean's List , University of Illinois at Urbana-Champaign	2024, 2025
First Prize Scholarship for Academic Excellence , ZJUI	2025
First Prize , China Southeast Mathematical Olympiad (CSMO)	2021, 2022

Selective Coursework

◦ CS 225: Data Structures (A+)	◦ CS 411: Database Systems (A)
◦ ECE 391: Computer Systems Engineering (A+)	◦ ECE 310: Digital Signal Processing (A+)
◦ ECE 313: Probability with Engrg Applic (A+)	◦ ECE 498: Deep Generative Models (A+)

Teaching Experience

Teaching Assistant , ECE 220: Computer Systems & Programming	<i>Spring 2025</i>
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- Led office hours and discussion sessions for 300+ students; provided guidance on C programming and debugging
- Refactored and automated autograder scripts, reducing grading time and improving feedback efficiency

Teaching Assistant , MATH 257: Linear Algebra	<i>Fall 2024</i>
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- Conducted weekly problem-solving sessions for 100+ students; organized comprehensive final review session